

Projection to 2045: Economic Performance of MSW Recovery Pathways

Net cost, abatement cost and the preferred option for 21 Indonesian cities and regencies, 2045 against the existing condition of 2025, with real escalation of electricity (2%/yr), labour-driven O&M (1%/yr) and landfill cost (2%/yr) and a decarbonising grid — MSW pathway model v3.2

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Key findings

- Net cost of the **landfill** rises from 18.4 to 25.9 USD/t (median): landfill escalation outweighs the economies of scale of larger sites.
- **WtE becomes cheaper** (57.9 → 50.8 USD/t): electricity revenue grows by 49% and larger plants are cheaper per tonne. Locations reaching 1,000 t/day (Perpres 109/2025 tariff) rise from 6 to 8.
- **RDF, AD and RDF + AD** become dearer by about 5–8 USD/t, mainly through O&M and residue-landfill escalation.
- The **abatement cost** of RDF + AD against the landfill stays near 72 USD/t CO₂e (2025: 73); that of WtE stays near 98 because its lower cost is offset by its smaller climate benefit.
- **The preferred option hardly changes**: landfill up to 50 USD/t in both years; at 100 USD/t RDF + AD is most probable in 19 locations in 2045 (2025: 18). The decision rules of the main study remain valid under these drivers.

1 What is projected and how

The six options are evaluated per tonne of mixed MSW at the facility gate for the existing condition of 2025 and for 2045, for the same 21 locations, with the same model, parameters and composition. Three drivers change between the two years (Table 1); everything else is held constant so that the difference isolates their effect. Both years use the same Monte Carlo random numbers, so changes are computed draw by draw (paired).

$$EF_{grid}(t) = EF_{grid}(2025) \max[0, 1 - \delta(t - 2025)], \quad \delta = \frac{1}{35} \text{ yr}^{-1} \Rightarrow EF_{grid}(2045) = 0.429 EF_{grid}(2025) \quad (1)$$

$$x(t) = x(2025) (1 + e_x)^{t - 2025}, \quad Q(t) = Q(2025) (1 + g)^{t - 2025} \quad (2)$$

Driver	Rule	Factor 2025 to 2045	Affects
Grid emission factor	linear to zero in 2060 ($\delta = 1/35$ per year)	$\times 0.429$	electricity credits of WtE and AD, electricity use of RDF and front-end sorting, fossil CED of grid electricity
Electricity value	real escalation 2%/yr	$\times 1.486$	revenue of WtE and AD
Labour-driven O&M	real escalation 1%/yr	$\times 1.220$	O&M of WtE, RDF and AD plants
Landfill cost	real escalation 2%/yr (third rate, see note)	$\times 1.486$	sanitary landfill and open-dump cost, including residue landfills of every option
Tonnage	each location's growth rate; composition constant	median $\times 1.41$	plant scale (economies of scale) and scale gates

Table 1: Drivers of the projection (data/projection_2045_parameters.csv). CAPEX is constant in real terms. The third escalation rate of the 2/1/2% set is applied to landfill cost, an interpretation to be confirmed. Grid factors in 2045: Jamali 0.877 → 0.376, Sumatera 0.832 → 0.357, Mahakam 1.128 → 0.483 kg CO₂/kWh.

All three drivers change costs: escalation raises O&M, landfill cost and electricity revenue; tonnage growth lowers the unit cost of capital through economies of scale (exponent $b = 0.7$) and of landfilling (elasticity e_{SL}); the grid affects x only indirectly, through the ranking by carbon-inclusive cost.

2 Net cost per tonne, 2025 and 2045

Option	C 2025	C 2045	Change	Change %	Escalation	Tonnage scale	Interaction
SL landfill + flare	18.4 (11.8 to 40.0)	25.9 (15.1 to 53.7)	+6.6	+34%	+9.0	-1.8	-0.9
S1 WtE	57.9 (42.4 to 120.3)	50.8 (34.8 to 107.1)	-8.1	-13%	-2.3	-5.4	-0.3
S2 RDF	42.3 (31.3 to 70.2)	49.8 (36.9 to 81.2)	+7.8	+18%	+10.6	-2.4	-0.6
S3 AD	36.6 (28.1 to 65.8)	43.0 (30.7 to 76.8)	+5.6	+15%	+8.8	-2.5	-0.7
S4 PHB	38.1 (22.6 to 75.1)	43.2 (24.2 to 86.5)	+5.0	+13%	+9.0	-3.4	-0.9
S5 RDF + AD	47.1 (37.3 to 78.2)	52.7 (41.0 to 85.9)	+5.5	+12%	+8.7	-2.7	-0.5

Table 2: Net levelised cost per tonne (USD/t, real 2025 USD), central values, market case: median (range) over 21 locations, and the median change attributed to each driver applied alone.

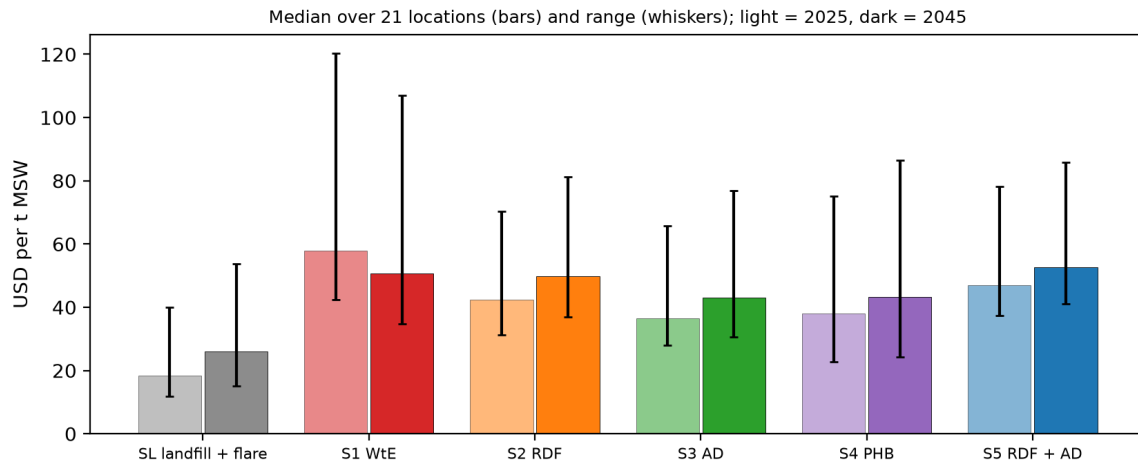


Figure 1: Net cost per tonne in 2025 and 2045.

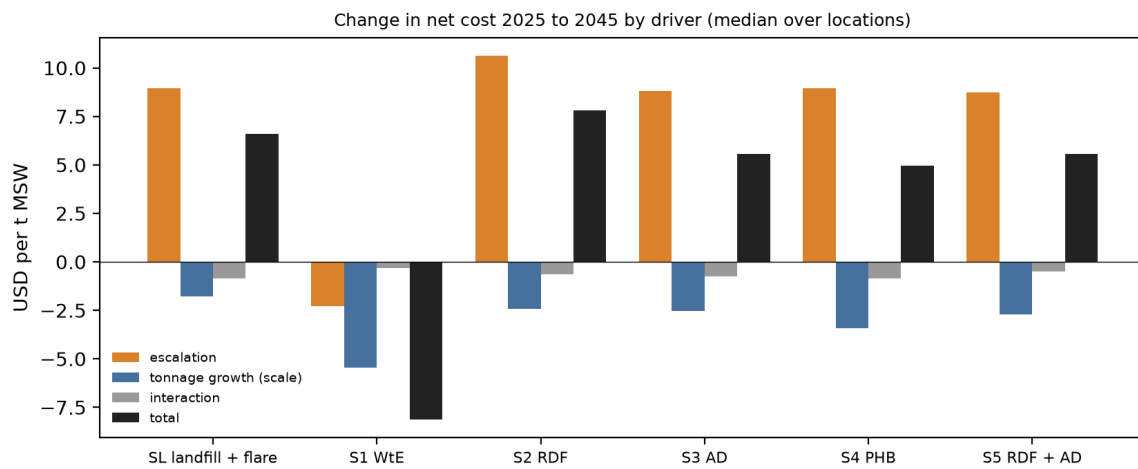


Figure 2: Decomposition of the change in net cost.

Option	Change p5	Change median	Change p95
SL landfill + flare	+4.0	+6.4	+8.9
S1 WtE	-16.3	-9.0	-3.8
S2 RDF	+5.7	+7.8	+9.9
S3 AD	+3.5	+5.7	+8.0
S4 PHB	+1.8	+4.6	+7.1
S5 RDF + AD	+3.8	+5.7	+7.6

Table 3: Uncertainty of the change in net cost (paired Monte Carlo, USD/t, median over locations).

3 Abatement cost and carbon-inclusive cost

Option	MAC vs SL 2025	MAC vs SL 2045	CIC at 50: 2025	CIC at 50: 2045	CIC at 100: 2025	CIC at 100: 2045
SL landfill + flare	-	-	45.8	53.0	72.7	78.9
S1 WtE	98	98	64.4	64.1	70.1	80.2
S2 RDF	98	93	56.0	64.5	72.3	78.3
S3 AD	112	117	56.0	63.9	75.6	83.4
S4 PHB	1449	1347	65.1	70.7	90.4	95.0
S5 RDF + AD	73	72	54.0	59.8	60.9	66.9

Table 4: Abatement cost against the landfill (USD/t CO₂e) and carbon-inclusive cost C + pG/1000 (USD/t) at 50 and 100 USD/t CO₂e, medians over 21 locations.

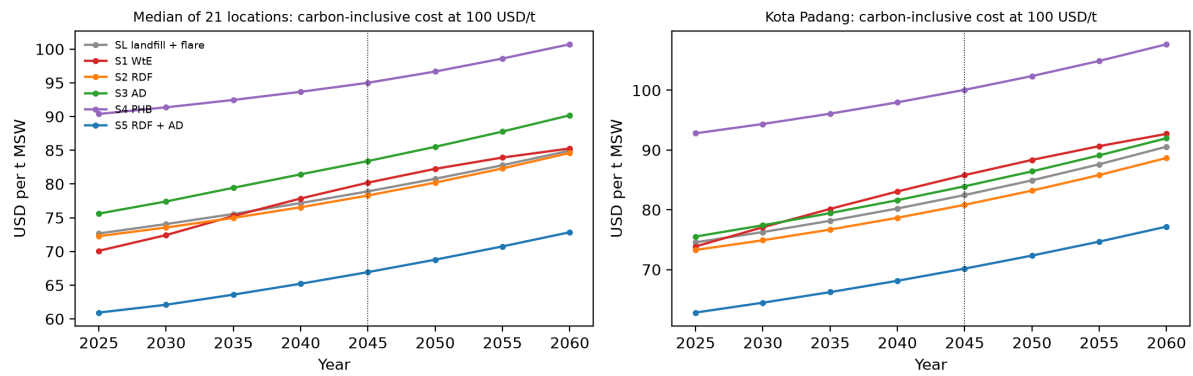


Figure 3: Carbon-inclusive cost at 100 USD/t CO_2e from 2025 to 2060 (central values, all drivers).

4 Preferred option at fixed carbon values

Carbon value (USD/t CO_2e)	Most probable 2025 (locations)	Most probable 2045 (locations)	Central best 2025	Central best 2045
0	SL 21	SL 21	SL 21	SL 21
2	SL 21	SL 21	as at 0	as at 0
25	SL 21	SL 21	SL 21	SL 21
50	SL 21	SL 21	SL 21	SL 21
100	S1 1, S5 18, SL 2	S5 19, SL 2	S1 1, S5 20	S1 1, S5 20

Table 5: Preferred option, 21 locations, market case (Monte Carlo 4,000 draws per location and carbon value; central values for the last two columns).

Location	Q 2025 t/d	Q 2045 t/d	P(S5) at 100: 2025	P(S5) at 100: 2045	P(SL) at 100: 2025	P(SL) at 100: 2045	Central best at 100: 2025 / 2045
Kota Padang	656	747	0.65	0.68	0.21	0.20	S5 / S5
Kab. Bogor	2,891	4,775	0.50	0.49	0.21	0.22	S5 / S5
Kab. Serang	1,187	1,696	0.32	0.36	0.37	0.37	S5 / S5
Kota Denpasar	918	1,302	0.47	0.49	0.07	0.06	S5 / S5
Kab. Cianjur	1,114	1,557	0.59	0.61	0.20	0.20	S5 / S5
Kab. Indramayu	1,152	1,442	0.62	0.63	0.21	0.22	S5 / S5
Kota Semarang	1,190	1,677	0.54	0.58	0.20	0.20	S5 / S5
Kab. Brebes	1,033	1,456	0.60	0.64	0.29	0.27	S5 / S5
Kab. Nganjuk	744	1,048	0.60	0.61	0.20	0.19	S5 / S5
Kab. Pati	691	974	0.83	0.85	0.08	0.06	S5 / S5
Kab. Magelang	671	946	0.62	0.65	0.19	0.18	S5 / S5
Kab. Pandeglang	527	742	0.41	0.45	0.36	0.34	S5 / S5
Kab. Pemasang	484	681	0.62	0.67	0.31	0.27	S5 / S5
Kab. Temanggung	423	596	0.41	0.46	0.48	0.44	S5 / S5
Kab. Pekalongan	560	789	0.66	0.71	0.12	0.10	S5 / S5
Kab. Sragen	402	567	0.62	0.67	0.20	0.17	S5 / S5
Kab. Kutai Kartanegara	373	526	0.00	0.00	0.45	0.48	S1 / S1
Kab. Kendal	437	616	0.67	0.68	0.20	0.18	S5 / S5
Kab. Wonogiri	422	595	0.70	0.73	0.15	0.14	S5 / S5
Kota Pekalongan	164	231	0.59	0.66	0.27	0.21	S5 / S5
Kota Magelang	50	70	0.50	0.65	0.37	0.24	S5 / S5

Table 6: Per-location results at 100 USD/t CO_2e (standard error of each probability ≤ 0.008).

5 Interpretation and limits

- Two effects cancel for WtE: higher electricity revenue and larger plants make it cheaper, while the cleaner grid removes much of its climate credit. Its position relative to RDF + AD therefore changes little, and it remains dependent on the tariff and on scale.
- Rising landfill costs make every option that reduces landfilling relatively more attractive, but by 2045 the effect is a few USD per tonne, smaller than the gap that the carbon value has to close.
- The 1,000 t/day gate of Perpres 109/2025 is reached by more locations as waste grows; whether that waste is collected is a separate question (managed shares are far lower today).

- CAPEX is constant in real terms and technology learning is not modelled; learning would lower the cost of the newer options (AD, RDF + AD, PHB). The Perpres tariff is held constant in real terms.
- Escalation rates and the grid path are user-specified scenario assumptions; the third rate of the 2/1/2% set is applied to landfill cost (to be confirmed). The parameters are in data/projection_2045_parameters.csv and the analysis reruns with python run_projection_2045.py.